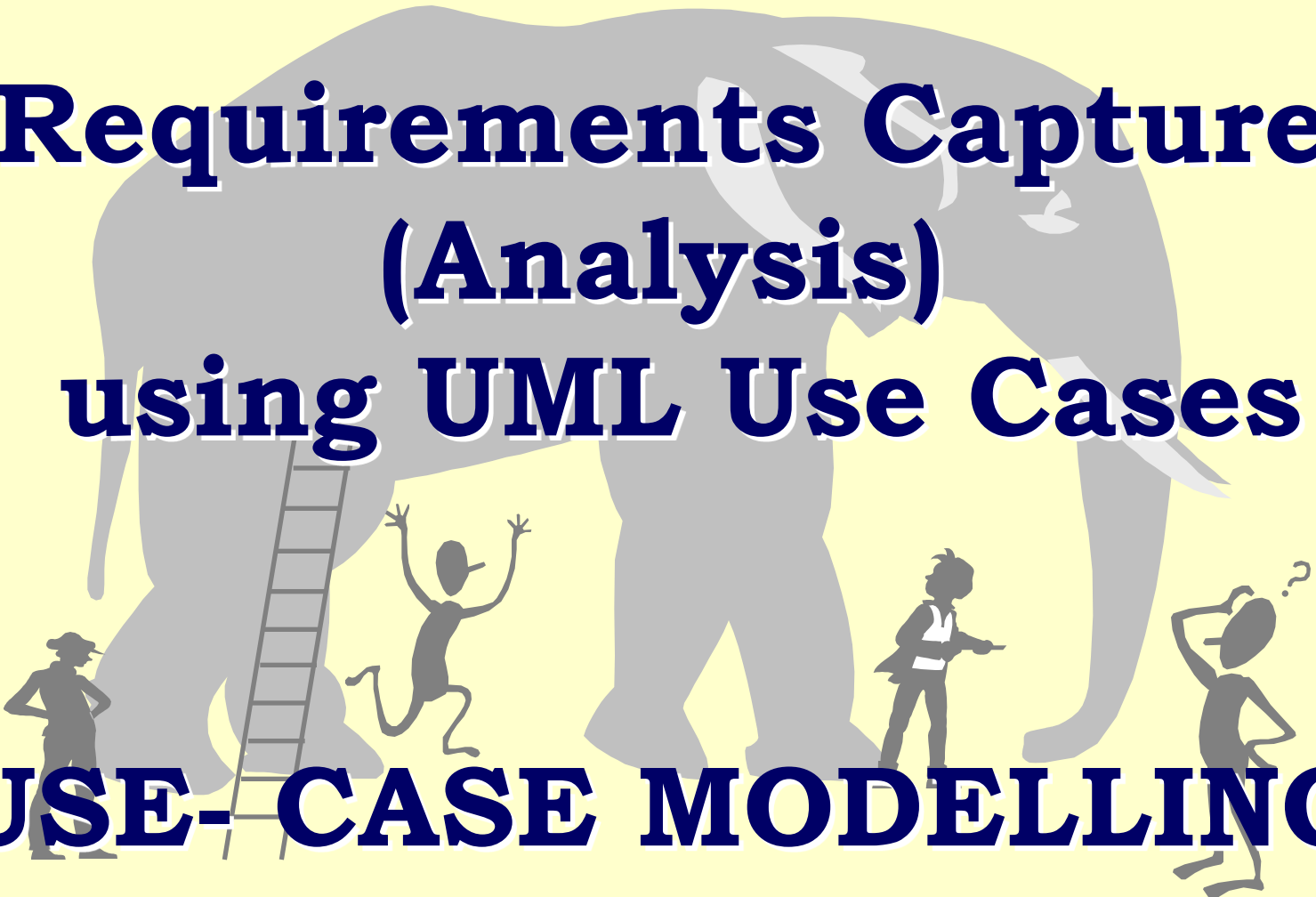


Requirements Capture (Analysis) using UML Use Cases

A large, light gray elephant is the central background element. Four small, dark gray stick figures are positioned around the elephant. One figure on the left stands with hands on hips. A second figure is climbing a ladder leaning against the elephant's leg. A third figure stands to the right of the elephant, holding a clipboard. A fourth figure on the far right is scratching its head with a question mark above it.

USE-CASE MODELLING

Contents

- ◆ Requirements Capture
- ◆ Building Blocks of a Use Case Diagram
 - ☐ *Actors*
 - ☐ *Use Cases*
 - ☐ *Relationships between Actors*
 - ☐ *Relationships between Use Cases*
- ◆ Terminology
 - ☐ *Flow of Events*
 - ☐ *Scenarios*
- ◆ Use Case DOs and DON'Ts

Use Case Diagram

- ◆ A graphical representation of the Use Cases of a system, its actors, and the interaction between them.
- ◆ It depicts the system boundary.
- ◆ Diagram Model elements
 - ☐ *Actors*
 - ☐ *Use cases*
 - ☐ *Relationships*
 - between Actors and Use Cases
 - between Use Cases
 - between Actors

Use Case

- ◆ Sequence of *transactions in a system* whose task is to yield results of a *measurable value* to an individual *actor* of the system.
- ◆ Is an abstraction of a set of sequences that yield some functionality.
- ◆ Represents some user-visible function.
- ◆ Is always initiated by an “actor”. Describes the interaction between the actors and the system for a system function.

Notation:



Reserve Seat

Finding Use Cases

- ◆ What functions does the system perform?
- ◆ What functions do the “actors” require?
- ◆ What input/output does the system need?
- ◆ What **verbs** are used to describe the system?

☐ *The Reservation Clerk **makes a booking** using the system, based on the...*

☐ *The Airport Manager can **make an entry for a new flight**. He can also **modify flight details**, provided...*

Use Case - Example

Indian Railways provides for advance reservation on all long-distance travel. The passenger seeking reservation of berth or seats should purchase the tickets from Railway Reservation Offices or Authorised Travel Agency only. To make an advance booking, the passenger is expected to fill in a prescribed application form and submit it to the reservation counter with the appropriate amount. Advanced Reservations are made up to 60 days in advance for all trains, for all classes exclusive of the day of departure of trains. An individual can book only up to six passengers on one requisition form provided all passengers are for the same destination and for the same train.

Use Case - Example (contd.)

Indian Railways wishes to develop a ticketing and reservation system. This must support advance booking of tickets, cancellation of tickets and change of class of a ticket. All these are handled by a Reservation Clerk.

The system will also have a web-interface where users can register themselves and purchase tickets online. They can pay either by using their online banking account or by credit card or by debit card. Reservations made over the internet can only be cancelled across the counter.

The system will also have a querying facility that allows users to check train time-tables, fares and availability of tickets.

Use Case - Example (contd.)

Use Cases :

Make
Reservation

Cancel
Reservation

Modify Class

Print Ticket

Query
Timetable

Check Fare

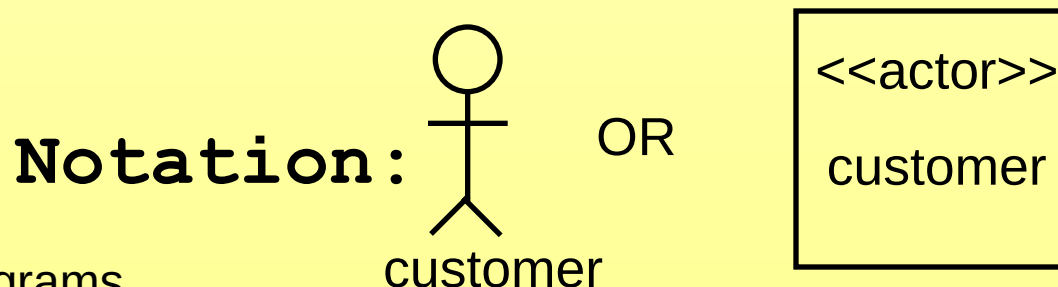
Register
as Member

Purpose of Use Cases

- ◆ To capture functional requirements of a system.
- ◆ To communicate with end users and domain experts.
- ◆ To design test cases for validating system functionality.
- ◆ To provide traceability from requirements into actual classes and operations.
- ◆ To drive the development process.
- ◆ To plan iterations and releases

Actors

- ◆ A role that interacts with the system.
- ◆ Represents a role, not individuals; can be a person or a device or another system.
- ◆ Communicate with the system by sending and/or receiving messages.
- ◆ An actor may participate in many use cases; a use case may have several actors participating in it.



Finding Actors

- ◆ Who uses the main functionality of the system?
- ◆ Who affects the system?
- ◆ Which hardware devices the system needs to handle?
- ◆ Which other systems does the system need to interact with?
- ◆ What nouns/subjects are used to describe the system?

☐ *The Reservation Clerk makes a booking using the system, based on the...*

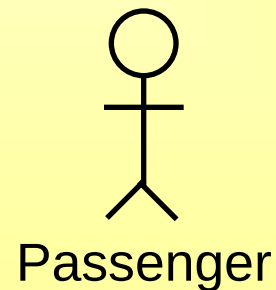
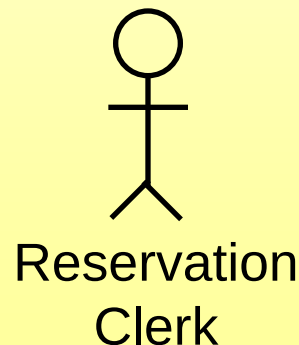
☐ *A user must login in order to save his itinerary*

Actors - Example

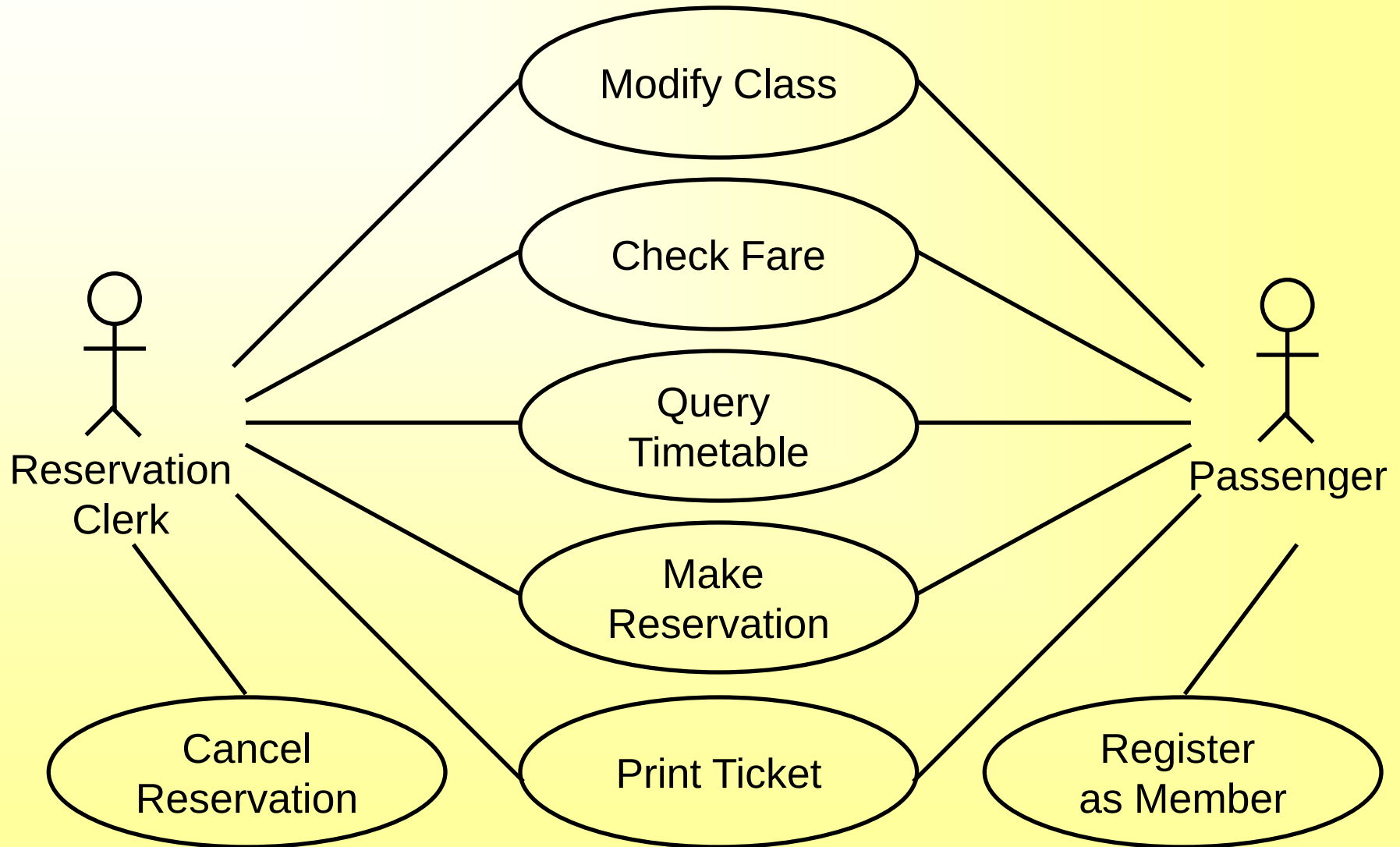
Indian Railways wishes to develop a ticketing and reservation system. This must support advance booking of tickets, cancellation of tickets and change of class of a ticket.

The system will also have a web-interface where users can register themselves and purchase tickets online.

Actors :



Use Case Diagram: Example



Relationships between Use Cases

- ◆ Uses/Include (Dependency)
- ◆ Extend (Dependency)
- ◆ Generalization
- ◆ Association (Between Actor and Use case)

“Extend” relationship : «extend»

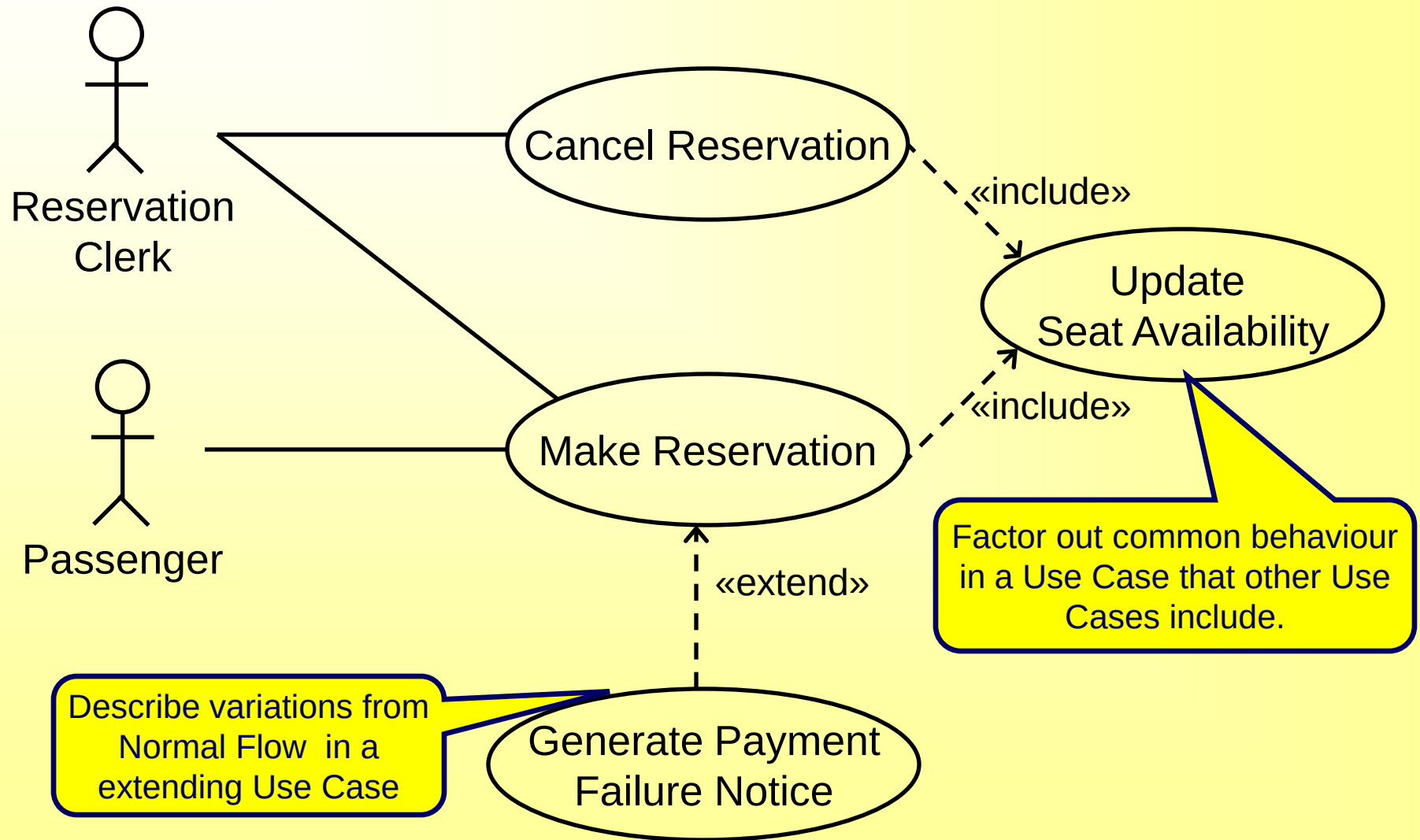
- ☐ *Use Case B extends Use Case A when Use Case B describes the behaviour of Use Case A under a particular condition.*
- ☐ *An extending Use Cases is used to describe variations in the normal flow of events described by a general use case*

Relationships between Use Cases

◆ “Include” relationship : «include»

- ☐ *Use Case A uses (or “uses”) Use Case B when Use Case B is a behaviour/functionality that is required by Use Case A. That behaviour has been factored out into a separate Use Case because it is required across several use cases.*
- ☐ *Common behavior in several use cases can be factored out into a single use case that is used by the other use cases*

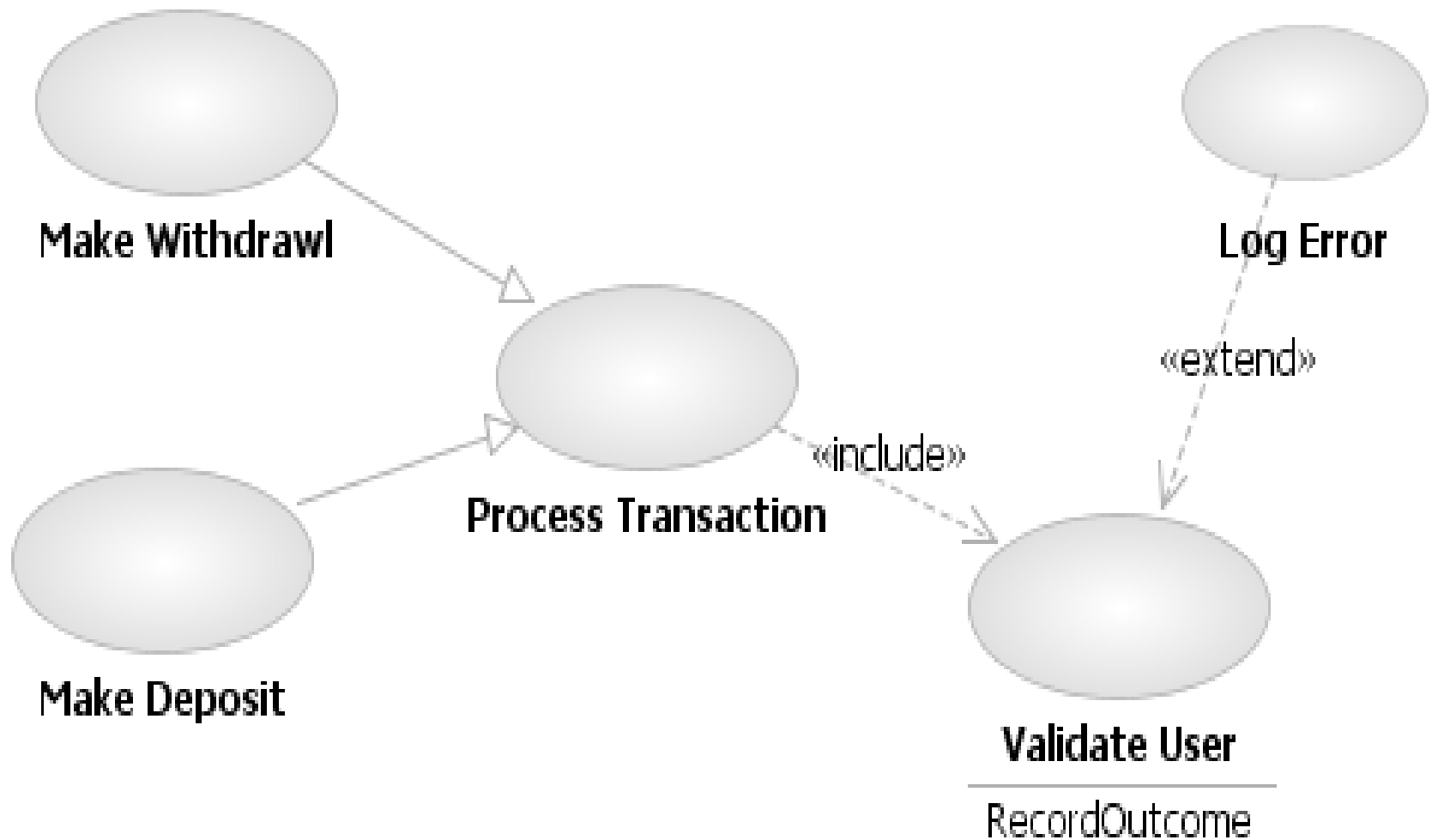
«extend» and «include» : Example



Difference between *uses* and *extends*

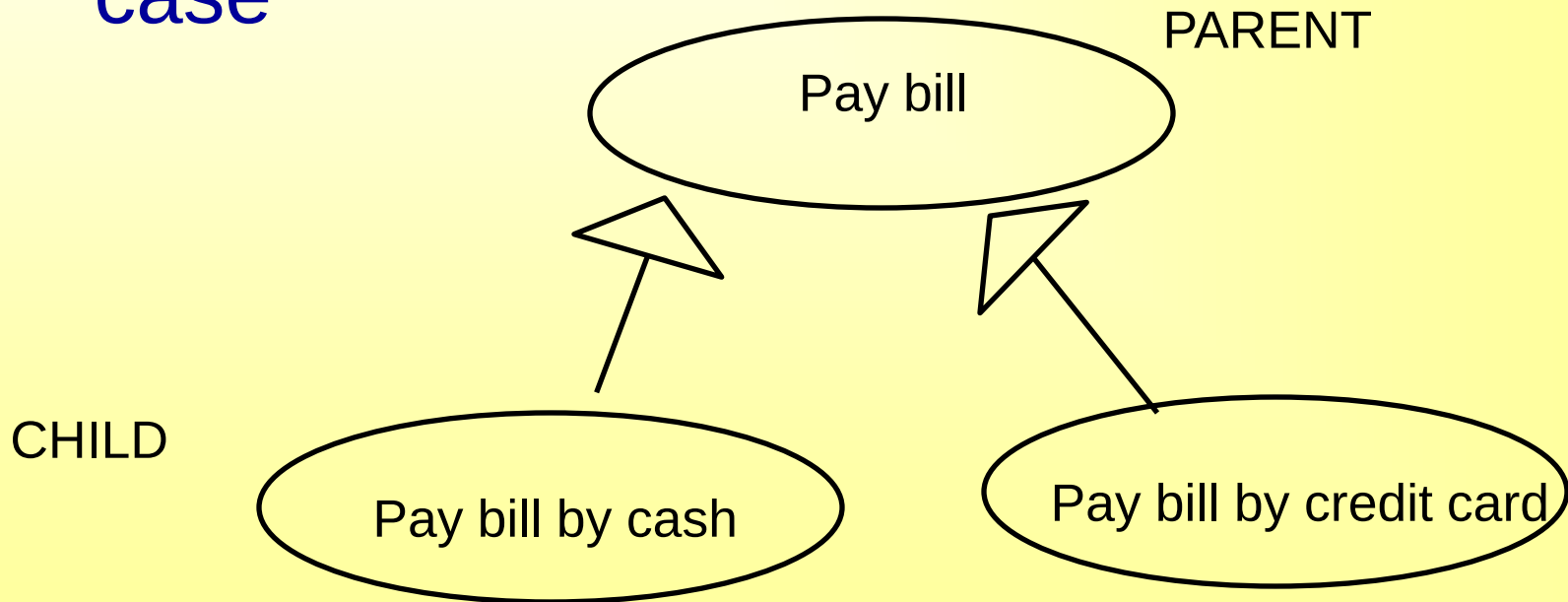
- ◆ Probably the best way to think about these diagram elements is as follows:
 - "*X uses Y*" indicates that the task "*X*" has a subtask "*Y*"; that is, in the process of completing task "*X*", task "*Y*" will be completed at least once.
 - "*X extends Y*" indicates that "*X*" is a task of the same type as "*Y*", but "*X*" is a special, more specific case of doing "*Y*". That is, doing *X* is a lot like doing *Y*, but *X* has a few extra processes to it that go above and beyond the things that must be done in order to complete *Y*.

Difference between include and extend



Generalization

- ◆ Parent – child relationship between use cases
- ◆ Child use case can replace parent use case



Use Case Description

- ◆ Is a text description of the use case functionality in the user language and terminology
- ◆ No specific UML format
- ◆ Describes WHAT and not HOW
- ◆ Typically includes:
 - ☐ *Objectives of the use case*
 - ☐ *How the use case is initiated*
 - ☐ *The flow of events*
 - ☐ *Alternate flow in the use case*
 - ☐ *How the use case finishes with a value to the actor****and more...***

Flow of Events

- ◆ Use Case is an abstraction of behaviour (set of sequences).
- ◆ The behaviour of the Use Case can be described by a “flow of events” - which spells out in detail what exactly the Use Case does.
- ◆ Flow of events specifies:
 - ☐ *the main flow of events (what happens and in what order when all is well).*
 - ☐ *alternate flow(s) of events (what happens and in what order when something goes wrong).*

Use Cases and Scenarios

- ◆ A Use Case actually describes a set of sequences [of actions].
- ◆ Each sequence represents one possible flow of actions in using the system.
- ◆ Each sequence is called a Scenario.
- ◆ A Scenario is basically one instance of a use case.

 *a Scenario is to a Use Case, what an Object is to a Class.*

Use Case Description - Example

Use Case: Make Reservation

Actors: Passenger, Reservation Clerk

Purpose: Reserve a seat or berth

Overview: Allows the user to make a reservation for a journey.

Normal Flow:

1. User logs in
2. User specifies the train and journey details.
3. User specifies passenger details
4. User specifies payment details
5. User confirms transaction

Actual Use Case Description will vary in structure as well as content.

Standard Template for Use Case Description

◆ Refer Document

Alternate Techniques to find Use Cases

- ◆ User goal technique
- ◆ CRUD technique
- ◆ Event Decomposition technique

User goal Technique

- ◆ An approach for identifying use cases in which an analyst talks to all users to get them to describe their goals in using the system.

User/actor	User goal and resulting use case
Order clerk	Look up item availability Create new order Update order
Shipping clerk	Record order fulfillment Record back order
Merchandising manager	Create special promotion Produce catalog activity report

CRUD Technique

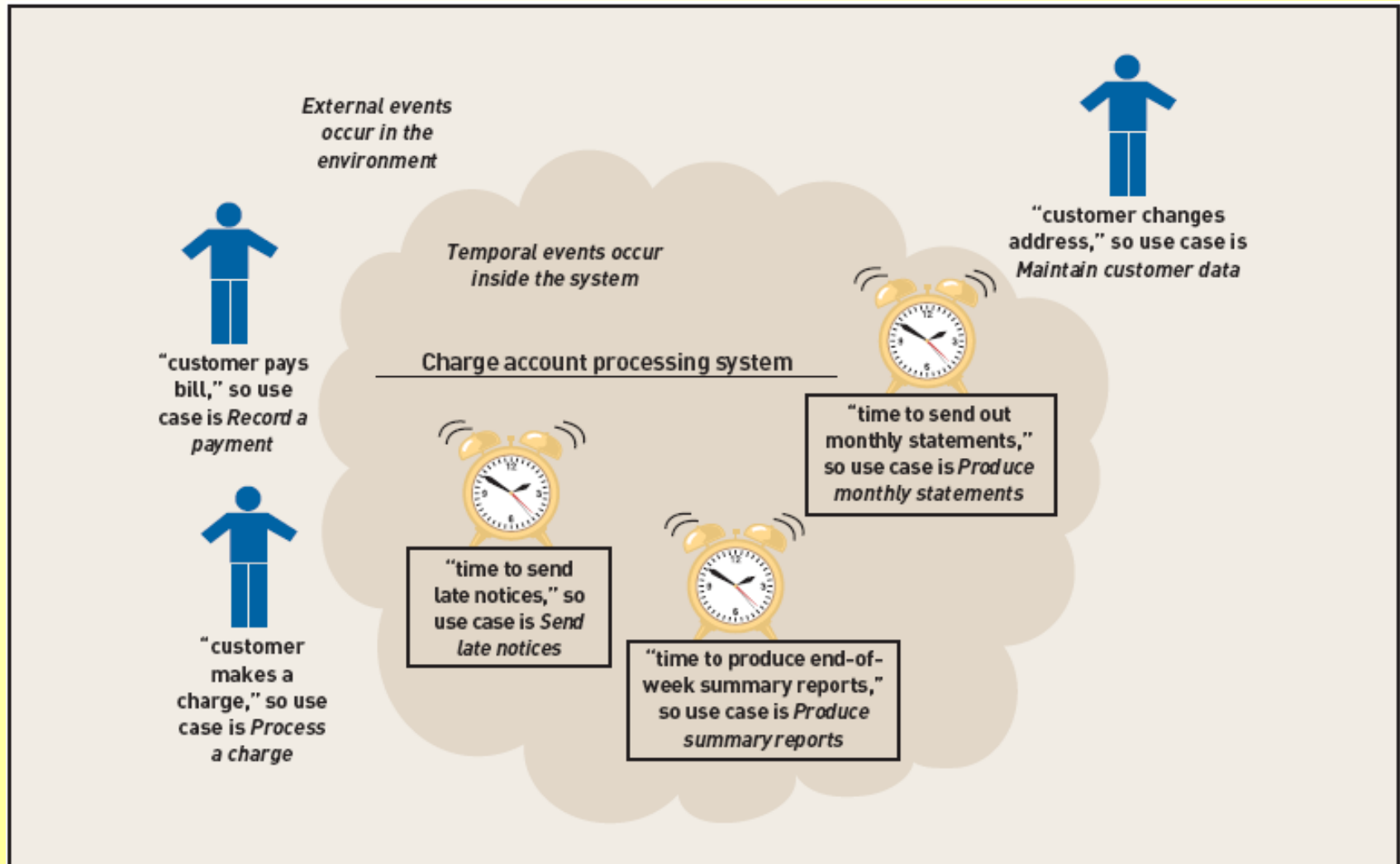
- ◆ An approach in which an analyst looks at each type of data and includes use cases that **create** the data, **read or report** on the data, **update** the data, and **delete** the data.

Data entity/class	CRUD	Resulting use case
Customer	Create	Add new customer
	Read/Report	Look up customer Produce customer list List customer orders
	Update	Update customer information
	Delete	Delete inactive customer

Event Decomposition Technique

- ◆ An analysis technique that focuses on identifying the events to which a system must respond and then determining how the system must respond.

Event Decomposition Technique



Steps in Use Case Modeling

- ◆ Establish the system boundary. Identify the actors that use the system.
- ◆ For each actor, consider what functions the system provides for each Actor.
- ◆ Represent each function as a Use case.
- ◆ Connect the Actor to all Use Cases initiated by the Actor.
- ◆ If the Use Case requires participation by other Actors, connect these actors also to the Use Case.

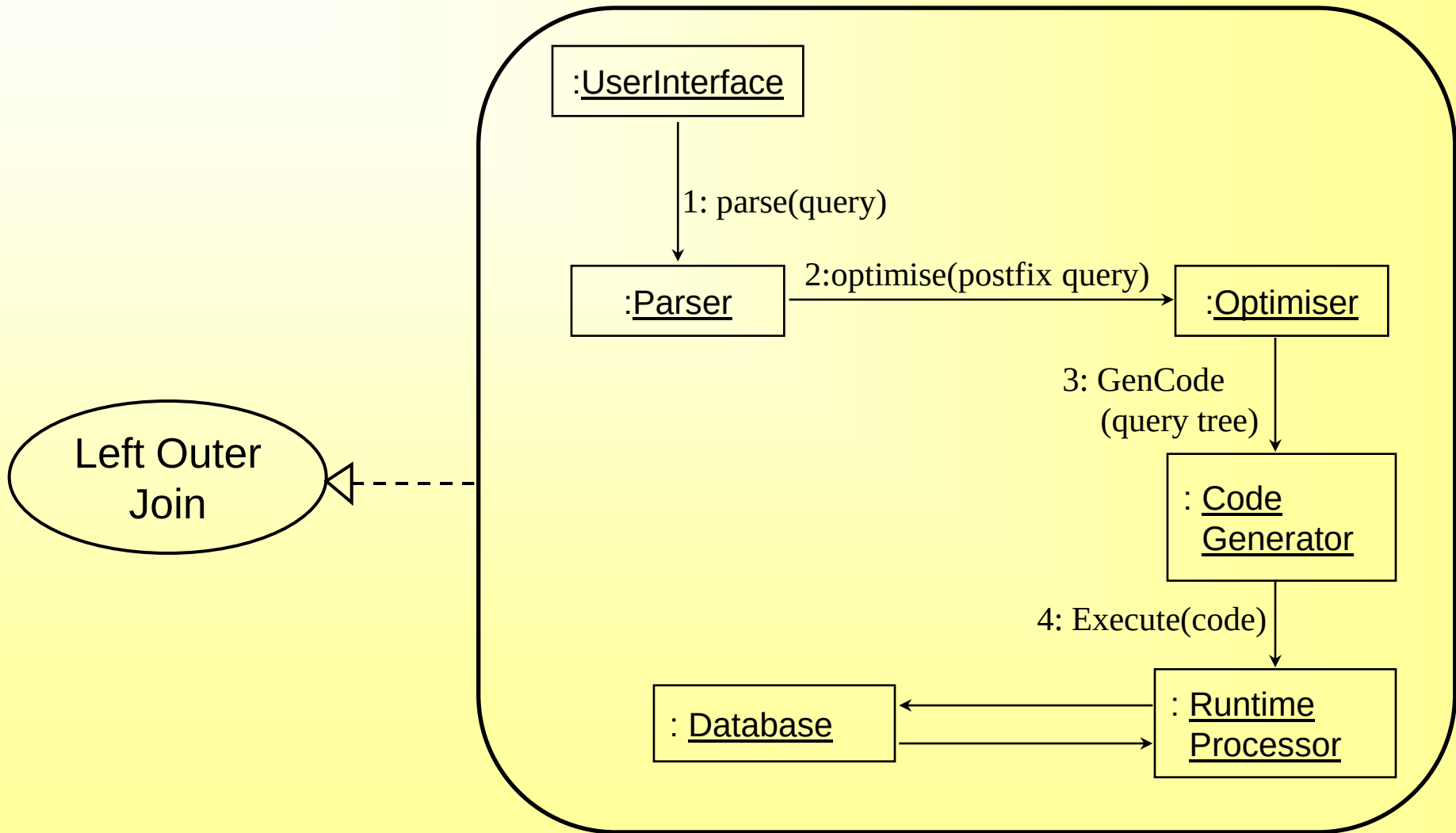
Steps in Use Case Modeling

- ◆ Factor optional or **exceptional behavior** in a Use Case, into a separate Use Case and connect this new Use Case with the main Use Case with the **«extend»** relationship.
- ◆ Analyze all Use Cases and identify **common functions** across groups of Use Cases. Factor each such common function into a separate Use Case and connect this common Use Case with the other Use Cases that use it, with the **«include»** relationship.

Use Case Realization

- ◆ The use case diagram is an external view of the system
- ◆ A use case is *realized* in a collaboration
- ◆ A collaboration shows an internal implementation- dependent solution of a use case in terms of:
 - ☐ *classes/objects*
 - ☐ *their relationships*
 - ☐ *their interaction*

Realization: Example



Courseware Management System

Problem Statement

- ◆ The organization offers a variety of courses in a variety of areas such as learning management techniques and understanding different software languages and technologies.
- ◆ Each course is made up of a set of topics.
- ◆ Tutors in the organization are assigned courses to teach according to the area that they specialize in and their availability.
- ◆ The organization publishes and maintains a calendar of the different courses and the assigned tutors every year.

Courseware Management System

- ◆ There is a group of course administrators in the organization who manage the courses including course content, assign courses to tutors, and define the course schedule.
- ◆ The training organization aims to use the Courseware Management System to get a better control and visibility to the management of courses as also to streamline the process of generating and managing the schedule of the different courses.

Identifying Actors

- ◆ Tutors
- ◆ Course administrators
- ◆ Students

Identifying Use Cases

- ◆ View courses
- ◆ Manage topics for a course
- ◆ Manage course information
- ◆ View course calendar
- ◆ View tutors
- ◆ Manage tutor information
- ◆ Assign courses to tutors

DRAW USE CASE DIAGRAM

DRAW USE CASE DIAGRAM FOR GIVEN SCENARIO

Read the following scenario:

The ViaNet bank client must be able to deposit an amount to and withdraw an amount from his or her accounts using the bank application. Each transaction must be recorded, and the client must have the ability to review all transaction performed against a given account. Recorded transactions must include the date, time, transaction type, amount , and account balance after the transaction.

A ViaNet bank client can have two types of accounts: a checking account and a savings account. For each checking account, one related saving account can exists.

DRAW USE CASE DIAGRAM FOR GIVEN SCENARIO

The application must verify that a client can gain access to his or her account by identification via personal identification number (PIN) code.

Neither a checking account nor a savings account can have a negative balance. The application should automatically withdraw funds from a related savings account if the requested withdrawal amount on checking account is more than its current balance.

If the savings account balance is insufficient to cover the requested withdrawal amount, the application should inform the user and terminate the transaction.

Univ. Questions

- ◆ What is use case diagram? Explain with notations and example.
- ◆ Draw Use Case diagram for TRAVEL AGENCY. [D-2010]
- ◆ Construct Use Case diagram for Online Airline Reservation System.[May -2011]
- ◆ For Hospital Management System, Draw Use Case Diagram.[May -2010]
- ◆ Construct Use Case diagram for Online Railway Reservation System.[D-2012]
- ◆ Draw use case diagram of Library Management System with include, extend and generalization relationships between use cases. [May 2016]

Univ. Questions

- ◆ Consider use case diagram for Course Registration System and explain extend, include and generalize relation between use cases.[Dec 14]
- ◆ Draw use case diagram for course registration system and write the use case specification for any two cases.[May 2015]
- ◆ Explain with an example, how use case modeling is used to describe functional requirements. Identify the actors, scenario and use cases for the example. [Dec 2015]
- ◆ Draw use case diagram for online railway reservation system with extend, include and generalize relations between use cases. [D-16]

Univ. Questions

- ◆ Explain the purpose of use case diagram with example.[M-2017]